14.5 Advanced configurations

Elasticsearch has many settings and configurations that may baffle even expert engineers. Although it uses the *convention over configuration* paradigm and works with defaults most of the time, it is imperative to customize the configuration before taking the application to production.

In this section, we walk through some properties that fall under various categories and discuss their importance and how to tweak them. We can modify three configuration files:

- *elasticsearch.yml*—The most commonly edited configuration file, where we can set the cluster name, node information, data and log paths, and network and security settings
- *log4j2.properties*—Where we can set the logging levels of the Elasticsearch node
- *jvm.options*—Where we can set the heap memory of the running node

These files are read by the Elasticsearch node from the config directory, which is a folder under Elasticsearch's installation directory. This directory defaults to $ES_HOME/config for binary (zip or tar.gz) installations (the `ES_HOME` variable points to Elasticsearch's installation directory). If we install with a package manager like Debian or RPM distributions, it defaults to /etc/elasticsearch/config.

If we expect to access our configuration files from a different directory, we can set and export a path variable called `ES_PATH_CONF`, which points to the new configuration file location. In the next few subsections, we look at some settings that are important for both administrators developers to understand.

14.5.1 The main configuration file

Although the folks at Elastic developed Elasticsearch to run with defaults (convention over configuration), we are highly unlikely to rely on defaults when taking the node into production. We should tweak the properties to set specific network information, data or log paths, security aspects, and so on. To do so, we can amend the elasticsearch.yml file to set most of the required properties for our running applications.

We briefly touched on networking properties earlier in this chapter when discussing node communication. Elasticsearch exposes network properties as `network.*` attributes. We can set the host names and port numbers using this property. For example, we can change the port number of Elasticsearch to 9900 rather than holding on to default port 9200: `http.port :9900`. We can also set `transport.port` to change the port on which nodes communicate internally.

You may need to alter many properties, depending on your requirements. Refer to the official documentation to find more about these properties in detail: http://mng.bz/yQNE.

14.5.2 Logging options

Elasticsearch is developed in Java, and like most Java applications, it uses Log4j 2 as its logging library. A running node spits out logging information at the `INFO` level to the console and a file (using Kibana Console and rolling file appenders, respectively).